Abbring & Van den Berg (2003), built up step by step

Plan

1. One duration: survival, hazard, integrated hazard
2. People differ: unobserved heterogeneity and dynamic selection
3. The mixed proportional hazard (MPH) model
4. Two durations racing: competing risks
5. Adding a treatment: potential outcomes indexed by treatment time, no anticipation
6. What we actually observe
7. The model of the paper (Model 1a)
8. Identification, step by step
9. Extensions: heterogeneous effects, multiple spells
10. Connections: DiD with staggered adoption, sequential exogeneity, traps

Running example (the paper's, Sections 1 and 2.4)

- Unemployed workers receive unemployment benefits.
- At a random time S during the spell, their **benefits are reduced** (from level w to $b < w$).
- Outcome: unemployment duration Y.
- Question: what is the causal effect of the benefits reduction on the **transition rate to work**?

Road map: why each section is needed

- The effect is on a *rate* ⇒ we need the language of rates (Section 1).
- Workers differ in ways we do not see, and those still unemployed are selected ⇒ Sections 2–3.
- Before treatment, two things compete: finding a job vs. the benefits reduction ⇒ Section 4.
- Define the causal effect and what we observe ⇒ Sections 5–6.
- Put it together and show the effect is identified ⇒ Sections 7–8.

1 One duration

→ *Why now:* the paper's effect is on a *transition rate*. Before anything else: what is a rate for one duration, and how does it relate to probabilities?

Objects

- $Y \geq 0$: a continuous duration; y: a value.
- Density $f(y)$, CDF $F(y)$.
- Survival function $S(y) := \Pr(Y > y) = 1 - F(y)$: share still in the state at y.

1.1 Hazard rate

→ *Why now:* this is the object the treatment effect will act on.

$$\begin{aligned}
\theta(y) &:= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy \mid Y \ge y)}{dy} && \text{[definition]} \\
&= \lim_{dy \downarrow 0} \frac{\Pr(y \le Y < y + dy)}{dy \ \Pr(Y \ge y)} && \text{[conditional probability: } \Pr(A \mid B) = \Pr(A \cap B)/\Pr(B)\text{]} \\
&= \lim_{dy \downarrow 0} \frac{F(y + dy) - F(y)}{dy} \cdot \frac{1}{S(y)} && \text{[}Y \text{ continuous: } \Pr(Y \ge y) = \Pr(Y > y)\text{]} \\
&= \frac{f(y)}{S(y)} && \text{[definition of the derivative: } F' = f\text{]} \qquad (1)
\end{aligned}$$

- Exit rate *among those still in the state* at y.
- $\theta(y)\, dy \approx$ probability of leaving in $[y, y + dy)$, given still there at y.
- A rate, not a probability: can exceed 1.

1.2 Integrated hazard and survival

→ *Why now:* we model hazards, but data and likelihoods are about probabilities. We need the map from θ back to S and f; it is used in every later derivation.

$$\begin{aligned}
\theta(y) &= \frac{f(y)}{S(y)} && \text{[by (1)]} \\
&= -\frac{S'(y)}{S(y)} && \text{[}S = 1 - F\text{, so } S' = -f\text{]} \\
&= -\frac{d}{dy} \log S(y) && \text{[chain rule: } (\log S)' = S'/S\text{]}
\end{aligned}$$

$$\begin{aligned}
\Theta(y) := \int_0^y \theta(u)\, du &= -\big[\log S(u)\big]_0^y && \text{[fundamental theorem of calculus]} \\
&= -\log S(y) && \text{[}S(0) = 1\text{, so } \log S(0) = 0\text{]} \\
\implies \quad S(y) &= e^{-\Theta(y)} && \text{[exponentiate]} \qquad (2) \\
f(y) &= \theta(y)\, e^{-\Theta(y)} && \text{[}f = \theta S \text{ by (1)]} \qquad (3)
\end{aligned}$$

- Θ: integrated hazard.
- θ, Θ, S, f: four descriptions of the same distribution.

1.3 Example: constant hazard

→ *Why now:* the simplest benchmark; it defines "duration dependence", which Section 2 will show is hard to read from data.

$$\theta(y) = c \quad \Longrightarrow \quad \Theta(y) = cy, \qquad S(y) = e^{-cy} \qquad \text{[integrate; then (2)]}$$

$$\Pr(Y > y + u \mid Y > y) = \frac{S(y+u)}{S(y)} \qquad \text{[conditional probability; } \{Y > y+u\} \subset \{Y > y\}\text{]}$$

$$= e^{-cu} \qquad \text{[plug in } S\text{]}$$

- Does not depend on y: **memoryless**.
- θ decreasing in y: **negative duration dependence**. Increasing: **positive**.

1.4 Writing Y as a function of a standard random variable

→ *Why now:* the paper defines potential outcomes this way (Section 5), and it shows that treatment timing contains pure chance (Section 7).

Assume $\theta > 0$, so Θ is strictly increasing and invertible. Define $E := \Theta(Y)$. For $e \geq 0$:

$$\begin{aligned}
\Pr(E > e) &= \Pr\big(\Theta(Y) > e\big) && \text{[definition of } E\text{]} \\
&= \Pr\big(Y > \Theta^{-1}(e)\big) && \text{[apply } \Theta^{-1}\text{, strictly increasing]} \\
&= S\big(\Theta^{-1}(e)\big) && \text{[definition of } S\text{]} \\
&= e^{-\Theta(\Theta^{-1}(e))} = e^{-e} && \text{[by (2)]} \\
\Longrightarrow \quad Y &= \Theta^{-1}(E), \qquad E \sim \text{Exp}(1) && \text{[invert the definition of } E\text{]} \qquad (4)
\end{aligned}$$

- $E \sim \text{Exp}(1)$ *whatever* θ is.
- Same idea as $F(Y) \sim U(0,1)$: indeed $E = -\log S(Y)$.
- Y = model part Θ applied to a standard random part E. Used in Sections 5 and 7.

Takeaway → next: with one homogeneous population, the hazard is directly read from data via $\theta = f/S$. Next: what if people differ in ways we do not observe?

2 Unobserved heterogeneity and dynamic selection

→ *Why now:* workers differ in unobserved ways (e.g. motivation). The paper compares treated and not-yet-treated workers *among those still unemployed*. So we must know what heterogeneity does to the group that is still there.

Model

- Individuals differ by an unobserved $V \geq 0$, $V \sim G$.
- Individual hazard: $\theta(y \mid V) = \lambda(y)\, V$. Baseline λ, integrated baseline $\Lambda(y) := \int_0^y \lambda$.
- So $\Theta(y \mid V) = \Lambda(y)V$ and, by (2), $\Pr(Y > y \mid V) = e^{-\Lambda(y)V}$.

2.1 Population survival and hazard

→ *Why now:* we only observe the population, not each type. Compute what the population hazard looks like.

$$
\begin{aligned}
\bar{S}(y) := \Pr(Y > y) &= \mathbb{E}\big[\Pr(Y > y \mid V)\big] && \text{[law of iterated expectations (LIE)]} \\
&= \mathbb{E}\big[e^{-\Lambda(y)V}\big] && \text{[by (2), conditional on } V\text{]} \\
\bar{\theta}(y) &= -\frac{\bar{S}'(y)}{\bar{S}(y)} && \text{[by (1) and } S' = -f\text{]} \\
&= \frac{\mathbb{E}\big[\lambda(y) V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} && \text{[differentiate under } \mathbb{E};\ \Lambda' = \lambda\text{]} \\
&= \lambda(y)\,\frac{\mathbb{E}\big[V e^{-\Lambda(y)V}\big]}{\mathbb{E}\big[e^{-\Lambda(y)V}\big]} && \text{[linearity: } \lambda(y) \text{ is not random]} \qquad (5)
\end{aligned}
$$

2.2 Who survives: the type distribution among survivors

→ *Why now:* interpret the ratio in (5): it is about *who is left.*

$$
\begin{aligned}
dG(v \mid Y > y) &= \frac{\Pr(Y > y \mid V = v)\, dG(v)}{\Pr(Y > y)} && \text{[Bayes' rule]} \\
&= \frac{e^{-\Lambda(y)v}\, dG(v)}{\mathbb{E}[e^{-\Lambda(y)V}]} && \text{[plug in the two survivals above]} \qquad (6) \\
\mathbb{E}[V \mid Y > y] &= \frac{\mathbb{E}[V e^{-\Lambda(y)V}]}{\mathbb{E}[e^{-\Lambda(y)V}]} && \text{[integrate } v \text{ against (6)]} \\
\implies \quad \bar{\theta}(y) &= \lambda(y)\,\mathbb{E}[V \mid Y > y] && \text{[compare with (5)]} \qquad (7)
\end{aligned}
$$

- Survivors are the population **reweighted by their survival probability** $e^{-\Lambda(y)v}$.
- High-v people have lower weights: they leave first.

2.3 A name for the object that keeps appearing: the Laplace transform

→ *Why now:* every formula above contains $\mathbb{E}[e^{-aV}]$ for some a. Naming it and learning its properties is what makes all identification arguments below work.

Definition

$$
\mathcal{L}(a) := \mathbb{E}\big[e^{-aV}\big] = \int_0^\infty e^{-av}\, dG(v), \qquad a \geq 0. \qquad (8)
$$

- Reading in our model: $a = \Lambda(y)$ is the accumulated baseline hazard.
  - e^{-av}: probability that a type-v person is still unemployed.
  - $\mathcal{L}(a)$: share of the whole population still unemployed. Indeed $\bar{S}(y) = \mathcal{L}(\Lambda(y))$.
- Same family as the moment generating function: $\mathcal{L}(a) = M_V(-a)$, $M_V(u) := \mathbb{E}[e^{uV}]$.
  - Always exists, since $V \geq 0$ gives $0 < e^{-aV} \leq 1$.

Examples

Two types: v_1 with probability p, v_2 with probability $1 - p$:

$$
\mathcal{L}(a) = p\, e^{-av_1} + (1-p)\, e^{-av_2} \qquad \text{[expectation of a discrete r.v.]}
$$

Gamma with mean 1 and variance $1/k$:

$$\begin{aligned}
\mathcal{L}(a) &= \int_0^\infty e^{-av}\,\frac{k^k v^{k-1} e^{-kv}}{\Gamma(k)}\,dv && \text{[definition (8)]}\\
&= \frac{k^k}{(k+a)^k}\int_0^\infty \frac{(k+a)^k v^{k-1} e^{-(k+a)v}}{\Gamma(k)}\,dv && \text{[multiply and divide by } (k+a)^k\text{]}\\
&= \Big(1+\frac{a}{k}\Big)^{-k} && \text{[integrand is a Gamma}(k, k+a)\text{ density]}
\end{aligned}$$

Properties

$$\begin{aligned}
\mathcal{L}(0) &= \mathbb{E}[1] = 1 && \text{[definition]}\\
\mathcal{L}'(a) &= -\mathbb{E}\big[Ve^{-aV}\big] < 0 && \text{[differentiate under } \mathbb{E}\text{; } <0 \text{ if } \Pr(V>0)>0\text{]}\\
\mathcal{L}^{(k)}(a) &= (-1)^k\,\mathbb{E}\big[V^k e^{-aV}\big] && \text{[repeat } k \text{ times]}\\
-\mathcal{L}'(0) &= \mathbb{E}[V] && \text{[set } a=0\text{: derivatives at 0 give moments]}\\
-\frac{\mathcal{L}'(a)}{\mathcal{L}(a)} &= \frac{\mathbb{E}[Ve^{-aV}]}{\mathbb{E}[e^{-aV}]} = \mathbb{E}[V \mid \text{survived to } a] && \text{[by (6)]} \qquad (9)\\
\implies \quad \bar\theta(y) &= \lambda(y)\Big(-\frac{\mathcal{L}'}{\mathcal{L}}\Big)\big(\Lambda(y)\big) && \text{[by (7)]}
\end{aligned}$$

Two facts we state without proof:

- **Uniqueness:** $\mathcal{L}$ determines G. Two distributions with the same $\mathcal{L}$ are equal.
- **Analyticity:** for $a > 0$, $\mathcal{L}$ equals its Taylor series.
  - So knowing $\mathcal{L}$ on *any small interval* pins it down on all of $(0, \infty)$.
  - Like a polynomial: a short stretch of its graph fixes the whole graph.

Why this matters for identification

- Survival data are always $\mathcal{L}$ evaluated at some index (here $\Lambda(y)$).
- Strategy throughout: learn the index; learn $\mathcal{L}$ on an interval; then analyticity + uniqueness give G.

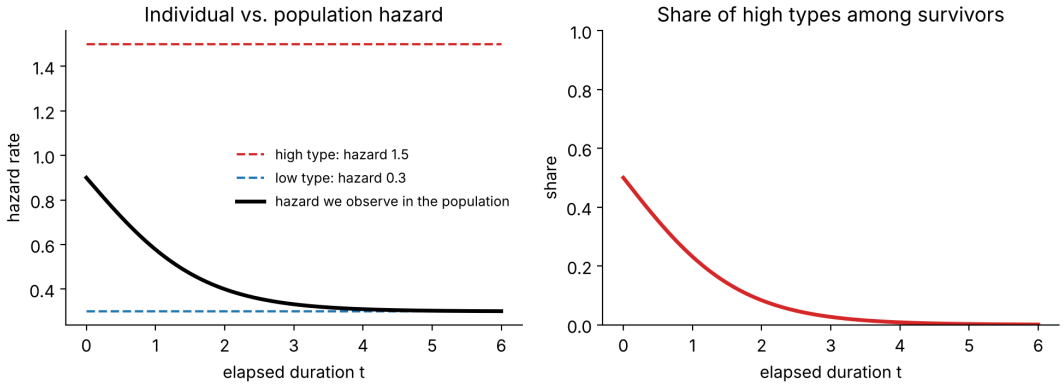
2.4 The mean type among survivors falls

→ *Why now:* show the *direction* of the composition effect in (7).

$$\begin{aligned}
\mathbb{E}[V \mid Y>y] &= -\frac{d}{da}\log\mathcal{L}(a)\Big|_{a=\Lambda(y)} && \text{[(9); chain rule: } (\log\mathcal{L})' = \mathcal{L}'/\mathcal{L}\text{]}\\
\frac{d^2}{da^2}\log\mathcal{L}(a) &= \frac{\mathcal{L}''}{\mathcal{L}} - \Big(\frac{\mathcal{L}'}{\mathcal{L}}\Big)^2 && \text{[quotient rule]}\\
&= \mathbb{E}[V^2 \mid \cdot] - \mathbb{E}[V \mid \cdot]^2 = \mathrm{Var}(V \mid \text{survived to } a) && \text{[properties above: survivor moments]}\\
\frac{d}{dy}\mathbb{E}[V \mid Y>y] &= -\lambda(y)\ \mathrm{Var}(V \mid Y>y)\ \le\ 0 && \text{[chain rule: } \Lambda' = \lambda\text{]}
\end{aligned}$$

- The population hazard falls *even if no individual hazard falls.* This is **dynamic selection**.
- Hence two explanations for a falling exit rate:
  - true duration dependence: $\lambda\downarrow$ (e.g. skill loss: a *trap*);

  - composition: $\mathbb{E}[V \mid Y > y] \downarrow$ (sorting on permanent heterogeneity).
- People still unemployed at month 6 are a selected group. Every comparison made at month 6 inherits this.



Two types with constant hazards 0.3 and 1.5 (equal shares). The population hazard falls from 0.9 to 0.3.

Takeaway → next: the observed hazard mixes λ (true duration dependence) and G (who is left). Next: can data separate them? Only with structure, which is the MPH model.

3 The mixed proportional hazard (MPH) model

→ *Why now:* Section 2 showed that λ and G are confounded in the observed hazard. Here: what structure lets data separate them? This is the one-duration version of the argument the paper runs with two durations (Section 4).

Model

- $\theta(y \mid x, V) = \lambda(y)\,\varphi(x)\,V$, with $V \sim G$ and $V \perp X$.
- Three multiplicative pieces: duration (λ), observed (φ), unobserved (V).
- Normalize $\varphi(x^*) = 1$ and $\Lambda(y^*) = 1$ for some chosen x^*, y^*.
  - Reason: (λ, φ, V) and $(c\lambda, \varphi, V/c)$ give the same hazard.

3.1 Survival through the Laplace transform

→ *Why now:* write what the data reveal, $S(y \mid x)$ and $\theta(y \mid x)$, in terms of the unknowns (λ, φ, G).

$$
\begin{aligned}
S(y \mid x) &= \mathbb{E}\big[e^{-\varphi(x)\Lambda(y)V}\big] && \text{[LIE over } V\text{; (2); } V \perp X\text{]} \\
&= \mathcal{L}\big(\varphi(x)\Lambda(y)\big) && \text{[definition (8)]} \qquad (10) \\
\theta(y \mid x) &= \lambda(y)\varphi(x)\Big(-\frac{\mathcal{L}'}{\mathcal{L}}\Big)\big(\varphi(x)\Lambda(y)\big) && \text{[}-\frac{d}{dy}\text{ log of (10); chain rule]} \qquad (11)
\end{aligned}
$$

- Data reveal one function $\mathcal{L}$ of a **single index** $\varphi(x)\Lambda(y)$.
- x and y both move the index: this is what the three steps below exploit.

3.2 Identification (Elbers & Ridder 1982) in three steps

→ *Why now:* recover φ, then G, then λ from $S(y \mid x)$. Each step uses one property of $\mathcal{L}$ from Section 2.3. Section 4 repeats these steps in two dimensions.

Step 1: φ from short durations. Assume $\mathbb{E}V < \infty$.

$$\frac{\theta(y \mid x)}{\theta(y \mid x^*)} = \varphi(x)\,\frac{(-\mathcal{L}'/\mathcal{L})(\varphi(x)\Lambda(y))}{(-\mathcal{L}'/\mathcal{L})(\Lambda(y))} \qquad \text{[by (11); } \lambda(y) \text{ cancels; } \varphi(x^*)=1\text{]}$$

$$\xrightarrow{y\downarrow 0} \varphi(x)\,\frac{-\mathcal{L}'(0)/\mathcal{L}(0)}{-\mathcal{L}'(0)/\mathcal{L}(0)} = \varphi(x) \qquad \text{[}\Lambda(y)\to 0\text{; continuity; } -\mathcal{L}'(0)=\mathbb{E}V<\infty\text{]}$$

- At $y \approx 0$ nobody has been selected out yet, so the ratio is free of selection.
- Uses short durations, where data are thin.

Step 2: G at the normalization date.

$$S(y^* \mid x) = \mathcal{L}\big(\varphi(x)\big) \qquad \text{[(10) with } \Lambda(y^*)=1\text{]}$$

- $\varphi(x)$ known from step 1. Varying x traces $\mathcal{L}$ on an interval.
- Analytic ⇒ $\mathcal{L}$ known on $(0,\infty)$ ⇒ G known.

Step 3: baseline.

$$\Lambda(y) = \frac{\mathcal{L}^{-1}\big(S(y \mid x)\big)}{\varphi(x)} \qquad \text{[invert (10); } \mathcal{L} \text{ strictly decreasing]}$$

What was assumed: proportionality, $V \perp X$, $\mathbb{E}V < \infty$, variation in $\varphi(x)$.

Takeaway → next: with MPH structure and covariates, one duration reveals λ, φ and the full distribution of unobserved types. In the paper, *before* treatment, two durations compete: job vs. benefits reduction. Their link through unobserved types is exactly the selection problem. Next: extend the argument to two durations.

4 Two durations racing: competing risks

→ *Why now:* before any treatment, a worker either finds a job (Y) or has benefits reduced (S), whichever comes first. If the same unobserved traits drive both (e.g. low motivation ⇒ slow job finding *and* early reduction), the treated are selected. This section asks: can the dependence between the two be learned from the race alone? No treatment effect is involved yet.

Set-up

- Two latent durations: Y (exit to work) and S (benefits reduction).
- Only the **first** event is seen, plus which one it was.
- MPH for each, with **correlated** unobservables $V = (V_Y, V_S) \sim G$, $V \perp X$:

$$\theta_Y(t \mid x, V) = \lambda_Y(t)\varphi_Y(x)V_Y, \qquad \theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S.$$

- Given (x, V): $Y \perp S$. All dependence runs through V.
- Bivariate Laplace transform: $\mathcal{L}(a,b) := \mathbb{E}[e^{-aV_Y - bV_S}]$; partials $\mathcal{L}_a, \mathcal{L}_b, \mathcal{L}_{ab}$.
  - Same properties as Section 2.3: derivatives give moments among survivors; uniqueness; analyticity.
- Indices: $a_t(x) := \varphi_Y(x)\Lambda_Y(t)$, $b_t(x) := \varphi_S(x)\Lambda_S(t)$.

4.1 Joint survival

→ *Why now:* the model's implication for the two latent durations together.

$$\begin{aligned}
\Pr(Y > y, S > s \mid x) &= \mathbb{E}\big[\Pr(Y > y, S > s \mid x, V)\big] && \text{[LIE over } V\text{; } V \perp X\text{]} \\
&= \mathbb{E}\big[\Pr(Y > y \mid x, V)\Pr(S > s \mid x, V)\big] && \text{[} Y \perp S \mid (x, V)\text{]} \\
&= \mathbb{E}\big[e^{-\varphi_Y(x)\Lambda_Y(y)V_Y}\, e^{-\varphi_S(x)\Lambda_S(s)V_S}\big] && \text{[(2) for each]} \\
&= \mathcal{L}\big(\varphi_Y(x)\Lambda_Y(y),\, \varphi_S(x)\Lambda_S(s)\big) && \text{[definition of } \mathcal{L}\text{]} \qquad (12)
\end{aligned}$$

4.2 What the data reveal

→ *Why now:* we never see both durations, only the first event. Derive which pieces of (12) are observable.

Probability that nothing has happened by t (the diagonal $y = s = t$):

$$P_0(t \mid x) := \Pr(Y > t, S > t \mid x) = \mathcal{L}\big(a_t(x), b_t(x)\big) \qquad \text{[(12) at } y = s = t\text{]} \qquad (13)$$

Cause-specific hazard of exit (rate of Y happening first, among those with nothing yet):

$$\begin{aligned}
h_Y(t \mid x) &:= \lim_{dt \downarrow 0} \frac{\Pr(t \le Y < t + dt,\ S > t \mid x)}{dt\ P_0(t \mid x)} && \text{[definition]} \\
&= \frac{-\partial_y \Pr(Y > y, S > t \mid x)\big|_{y=t}}{P_0(t \mid x)} && \text{[derivative of a survival in } y\text{]} \\
&= \frac{-\mathcal{L}_a(a_t, b_t)\,\varphi_Y(x)\lambda_Y(t)}{\mathcal{L}(a_t, b_t)} && \text{[chain rule on (12); } \Lambda_Y' = \lambda_Y\text{]} \\
&= \lambda_Y(t)\varphi_Y(x)\,\mathbb{E}[V_Y \mid R_t, x] && \text{[} -\mathcal{L}_a/\mathcal{L} = \text{survivor mean, as in (6)]} \qquad (14)
\end{aligned}$$

where $R_t := \{Y > t, S > t\}$ is the **risk set**. Symmetrically, $h_S(t \mid x) = \lambda_S(t)\varphi_S(x)\mathbb{E}[V_S \mid R_t, x]$.

4.3 Where the dependence of V_Y, V_S lives

→ *Why now:* the selection is the dependence of V_Y and V_S. Locate it inside the observable P_0.

Write $w(v) \propto e^{-av_Y - bv_S}$ (survivor weights) and $\mathbb{E}_w$ for expectations under them.

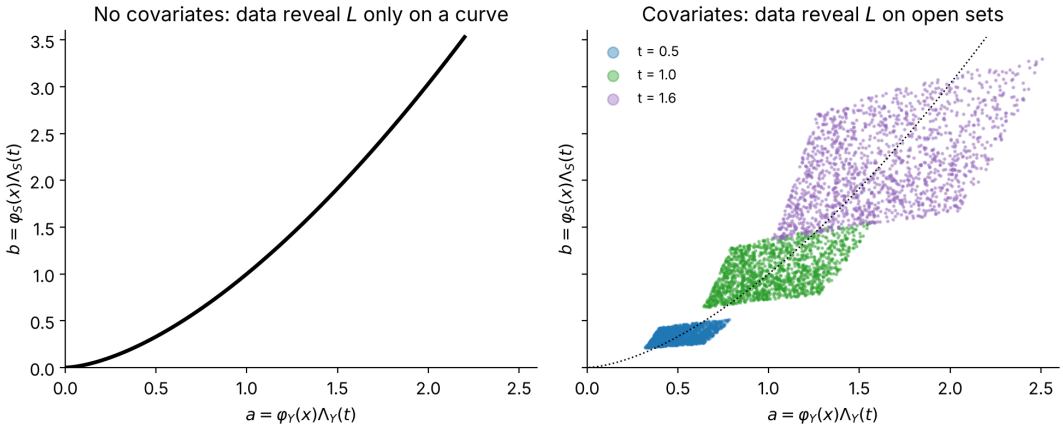
$$\begin{aligned}
\partial_a \log \mathcal{L} &= \frac{\mathcal{L}_a}{\mathcal{L}} = -\mathbb{E}_w[V_Y] && \text{[chain rule; differentiate under } \mathbb{E}\text{]} \\
\partial_b \partial_a \log \mathcal{L} &= \frac{\mathcal{L}_{ab}}{\mathcal{L}} - \frac{\mathcal{L}_a \mathcal{L}_b}{\mathcal{L}^2} && \text{[quotient rule]} \\
&= \mathbb{E}_w[V_Y V_S] - \mathbb{E}_w[V_Y]\mathbb{E}_w[V_S] && \text{[} \mathcal{L}_{ab} = \mathbb{E}[V_Y V_S e^{\cdots}],\ \mathcal{L}_b/\mathcal{L} = -\mathbb{E}_w[V_S]\text{]} \\
&= \mathrm{Cov}(V_Y, V_S \mid R_t, x) && \text{[definition of covariance; at } (a, b) = (a_t, b_t)\text{]} \qquad (15)
\end{aligned}$$

- $V_Y \perp V_S \iff \mathcal{L}(a, b) = \mathcal{L}_Y(a)\mathcal{L}_S(b) \iff \log P_0$ additively separable in the two indices.
- Dependence (selection) = an **interaction** of the two indices in $\log P_0$.

4.4 The problem: Tsiatis (1975)

→ *Why now:* the dependence sits in $\mathcal{L}$, but do the data reveal enough of $\mathcal{L}$? Without structure, no.

- Without structure, any competing-risks data can be generated by *independent* Y, S.
- Geometry, without covariates:
  - the data reveal $\mathcal{L}$ only at points $(\Lambda_Y(t), \Lambda_S(t))$, i.e. along a **curve**;
  - a curve cannot pin down a function of two variables, so the dependence is not identified.
- With covariates moving φ_Y and φ_S separately: points $(a_t(x), b_t(x))$ fill **open sets**.



4.5 Identification (Prop. 2) in three steps

→ *Why now:* show that MPH + covariates fix the Tsiatis problem: the steps of Section 3.2, in two dimensions.

Assumptions. (A3) $\{(\varphi_Y(x), \varphi_S(x))\}$ contains an open set of $\mathbb{R}^2$. (A4) $\mathbb{E}V_Y, \mathbb{E}V_S < \infty$. Normalize $\varphi_Y(x^*) = \varphi_S(x^*) = 1$, $\Lambda_Y(t^*) = \Lambda_S(t^*) = 1$.

Step 1: φ_Y, φ_S**.**

$$\frac{h_Y(t \mid x)}{h_Y(t \mid x^*)} = \varphi_Y(x)\,\frac{\mathbb{E}[V_Y \mid R_t, x]}{\mathbb{E}[V_Y \mid R_t, x^*]} \qquad \text{[by (14); } \lambda_Y(t) \text{ cancels]}$$

$$\xrightarrow{t\downarrow 0} \varphi_Y(x)\,\frac{\mathbb{E}V_Y}{\mathbb{E}V_Y} = \varphi_Y(x) \qquad \text{[risk set} \to \text{population; } V \perp X\text{; (A4)]}$$

Same for φ_S with h_S. Uses *which* event happened.

Step 2: G**, including the dependence.**

$$P_0(t^* \mid x) = \mathcal{L}\big(\varphi_Y(x), \varphi_S(x)\big) \qquad \text{[(13) with } \Lambda_Y(t^*) = \Lambda_S(t^*) = 1]$$

- Known points fill an open set (A3) ⇒ $\mathcal{L}$ known there.
- Analytic ⇒ known on $(0, \infty)^2$ ⇒ joint G known.

Step 3: baselines. For each t, $(\Lambda_Y(t), \Lambda_S(t))$ is the unique (a, b) with

$$P_0(t \mid x) = \mathcal{L}\big(\varphi_Y(x)a,\ \varphi_S(x)b\big) \quad \text{for all } x. \qquad \text{[(13); } \mathcal{L}, \varphi\text{'s known]}$$

- Uniqueness: two solutions would give $\mathcal{L}(u, v) = \mathcal{L}(ku, mv)$ on $(0, \infty)^2$, $(k, m) \neq (1, 1)$.

- Iterating, $\mathcal{L}$ would be constant, contradicting strict monotonicity.
- These three steps are our reconstruction; the paper cites Abbring & Van den Berg (2003, JRSSB).

Takeaway → next: the race alone reveals $\lambda_Y, \lambda_S, \varphi_Y, \varphi_S$ and the joint G, i.e. how selected those still at risk are. Next: introduce the treatment and define its causal effect.

5 Adding a treatment

→ *Why now:* so far no causal effect. We need to define "the effect of a benefits reduction at s", and say when the pre-treatment race of Section 4 is untouched by the treatment.

5.1 Potential outcomes indexed by treatment time

→ *Why now:* a dynamic treatment is a choice of *date*, so the causal objects are one outcome per date.

- Treatment = benefits reduction at date $s \in [0, \infty]$; $s = \infty$: never.
- One potential outcome per date: $Y^*(s)$, with hazard $\theta_{Y^*(s)}$, integrated hazard $\Theta_{Y^*(s)}$.
- By (4): $Y^*(s) = \Theta^{-1}_{Y^*(s)}(E)$, $E \sim \text{Exp}(1)$.
- Actual treatment date S; observed outcome $Y = Y^*(S)$.

5.2 No anticipation (Assumption 1)

→ *Why now:* for Section 4 to apply before treatment, the future treatment date must not affect the hazard before it.

$$\Theta_{Y^*(s)}(y) = \Theta_{Y^*(s')}(y) \qquad \text{for all } y \leq \min\{s, s'\}.$$

Take $s' = \infty$ and differentiate in y:

$$\theta_{Y^*(s)}(y) = \theta_{Y^*(\infty)}(y) \qquad \text{for } y < s. \qquad \text{[derivative of both sides]}$$

- Before treatment, everyone has the **untreated** hazard, whatever their future date.
- Economic content: people do not act on knowledge of their own future reduction date.
- Violated if, e.g., the date is announced in advance and people search harder before it.
- Not ruled out: reacting to the *policy* (e.g. the rate at which reductions happen).
  - This reaction is in every $Y^*(s)$.
  - So the effect is "treated at s vs. later, under this policy", not "policy vs. no policy".
- Prop. 1: without more structure, data cannot distinguish (i) causal effects from selection, nor (ii) anticipation from no anticipation. Hence Assumption 1 is maintained, and selection must be modelled.

Takeaway → next: under no anticipation, the period before treatment is a pure race, as in Section 4. Next: what exactly do the data contain?

6 What we actually observe

→ *Why now:* identification means recovering the model from the data distribution. We need that distribution precisely, including what is censored.

The censoring

- S is seen only if $S < Y$: if the worker finds a job first, the reduction date is never seen.
- Paper's data (per x): $Q_S(y,s) := \Pr(Y > y, S > s, Y > S)$ and $Q_Y(y) := \Pr(Y > y, Y < S)$. Assume $\Pr(Y = S) = 0$.

Pre-treatment objects

$$\begin{aligned}
P_0(t) &= \Pr(Y > t, S > t, Y > S) + \Pr(Y > t, S > t, Y < S) && \text{[partition by } Y \gtrless S;\ \Pr(Y=S)=0\text{]} \\
&= Q_S(t,t) + Q_Y(t) && [Y > t, Y < S \text{ implies } S > t] \\
h_Y(t) &= \frac{-Q_Y'(t)}{P_0(t)} && [-Q_Y' \text{: density of untreated exits}] \\
h_S(t) &= \frac{-\partial_s Q_S(y,s)\big|_{y\downarrow t,\, s=t}}{P_0(t)} && \text{[density of } S \text{ at } t\text{, still unemployed]}
\end{aligned}$$

Post-treatment object: survival of the cohort treated at s. For $y > s$:

$$\begin{aligned}
-\partial_s Q_S(y,s) &= \lim_{ds\downarrow 0} \frac{\Pr(Y > y,\ s \le S < s + ds,\ Y > S)}{ds} && \text{[derivative of } Q_S \text{ in } s] \\
&= \lim_{ds\downarrow 0} \frac{\Pr(Y > y,\ s \le S < s + ds)}{ds} && [Y > y > S \text{ implies } Y > S] \\
P_1(y \mid s) &:= \Pr(Y > y \mid S = s,\, Y > s) && \text{[definition]} \\
&= \frac{\partial_s Q_S(y,s)}{\lim_{y'\downarrow s} \partial_s Q_S(y',s)} && \text{[cond. probability; } ds \text{ cancels]} \qquad (16)
\end{aligned}$$

- (P_0, h_Y, h_S) = competing-risks data, as in Section 4.
- P_1 = the only place where the treatment effect can show up.

Takeaway → next: pre-treatment data identify the race (Section 4). Post-treatment data P_1 must deliver δ. Next: the model linking both.

7 The model of the paper (Model 1a)

→ *Why now:* combine the pieces: the competing-risks MPH model of Section 4 before treatment, and a causal effect δ after it.

Assumption 2 (selection on observables and unobservables): $Y^*(\cdot) \perp S \mid (X, V)$, with V time-invariant.

Model 1a

$$\theta_Y(t \mid s, x, V) = \begin{cases} \lambda_Y(t)\varphi_Y(x)V_Y & t \le s \\ \lambda_Y(t)\varphi_Y(x)V_Y\,\delta(t \mid s, x) & t > s \end{cases}$$
$$\theta_S(t \mid x, V) = \lambda_S(t)\varphi_S(x)V_S, \qquad (V_Y, V_S) \sim G, \quad V \perp X.$$

- δ: **the treatment effect**, a multiplier on the exit rate after treatment.
  - May depend on t, s (hence on time since treatment $t - s$), and x; not on V.
- No anticipation is built in: δ only acts for $t > s$.
- Selection on unobservables: V_Y and V_S may be dependent in any way.
- The same x in both hazards: **no exclusion restriction**.

- $\lambda_Y, \lambda_S, \varphi_Y, \varphi_S, G, \delta$ all nonparametric.
- Plus (A3), (A4) from Section 4.

Integrated outcome hazard, with $D(y \mid s, x) := \int_s^y \lambda_Y(\tau)\delta(\tau \mid s, x)\, d\tau$:

$$\begin{aligned}\Theta_Y(y \mid s, x, V) &= \int_0^s \lambda_Y \varphi_Y V_Y \, d\tau + \int_s^y \lambda_Y \varphi_Y V_Y \delta \, d\tau && \text{[split the integral at } s\text{]} \\ &= \varphi_Y(x) V_Y \big[\Lambda_Y(s) + D(y \mid s, x)\big] \quad (y > s) && \text{[linearity of the integral]} \end{aligned} \tag{17}$$

Treatment timing is random given (x, V)

$$E_S := \Theta_S(S \mid x, V) = \varphi_S(x) V_S \Lambda_S(S) \;\sim\; \text{Exp}(1), \quad E_S \perp (x, V) \qquad \text{[by (4), conditional on } (x, V)\text{]}$$

$$\implies \quad S = \Lambda_S^{-1}\Big(\frac{E_S}{\varphi_S(x) V_S}\Big) \qquad \text{[invert]}$$

- E_S moves S around without entering Y's hazard (Assumption 2).
- Like an instrument, except that what is excluded is an unobserved **shock**, not an observed variable.

Takeaway → next: the model has a random component in timing (like an instrument) and selection through V (to be learned from the race). Next: show δ is identified.

8 Identification, step by step

→ *Why now:* everything is in place. Plan:

- A: learn the selection (G) from pre-treatment data;
- B: describe who is treated at s;
- C: the simplest case, the moment of treatment;
- D: the general case.

8.1 Step A: the pre-treatment data are the competing-risks model of Section 4

→ *Why now:* check that the pre-treatment data are exactly the race of Section 4, so its results apply.

$$\begin{aligned} P_0(t \mid x) &= \mathbb{E}\big[\Pr(Y > t, S > t \mid x, V)\big] && \text{[LIE; } V \perp X\text{]} \\ &= \mathbb{E}\big[e^{-\varphi_Y(x)\Lambda_Y(t)V_Y}\, e^{-\varphi_S(x)\Lambda_S(t)V_S}\big] && \text{[Assumption 2; before } S\text{, } \delta \text{ absent (no anticipation)]} \\ &= \mathcal{L}\big(a_t(x), b_t(x)\big) && \text{[as in (13)]} \end{aligned}$$

- Same for h_Y, h_S. So Prop. 2 (Section 4.5) applies: $\varphi_Y, \varphi_S, \Lambda_Y, \Lambda_S, G$ are identified.
- This uses no post-treatment data.

8.2 Step B: who gets treated at s

→ *Why now:* the effect is estimated by following the cohort treated at s. We need its type distribution.

Risk set at s:

$$\begin{aligned} dG_{R_s}(v) &= \frac{\Pr(R_s \mid x, v)\, dG(v)}{\Pr(R_s \mid x)} && \text{[Bayes' rule]} \\ &\propto e^{-a_s v_Y - b_s v_S}\, dG(v) && \text{[as in Step A]} \end{aligned}$$

Cohort treated at s (in the risk set and treated in $[s, s+ds)$):

$$\begin{aligned} dG_s(v) &\propto \Pr(s \le S < s+ds \mid R_s, x, v)\, dG_{R_s}(v) && \text{[Bayes' rule]} \\ &\propto \lambda_S(s)\varphi_S(x)v_S\, ds\, dG_{R_s}(v) && \text{[definition of the hazard } \theta_S\text{]} \\ &\propto v_S\, e^{-a_s v_Y - b_s v_S}\, dG(v) && \text{[drop factors not depending on } v\text{]} \end{aligned} \tag{18}$$

- Treated cohort = risk set **reweighted by the treatment intensity** v_S.
- Fully known after Step A.

8.3 Step C: the selection problem at the moment of treatment

→ *Why now:* the cleanest comparison: right at s, compare the just-treated with those not yet treated. This isolates the selection problem in one factor.

Exit hazard of the just-treated vs. the not-yet-treated, at s:

$$\begin{aligned} \text{just treated:} \quad & \lambda_Y(s)\varphi_Y(x)\delta(s \mid s, x)\, \mathbb{E}_{G_s}[V_Y] && \text{[Model 1a, averaged over (18)]} \\ \text{not yet treated:} \quad & \lambda_Y(s)\varphi_Y(x)\, \mathbb{E}_{R_s}[V_Y] && \text{[(14)]} \end{aligned}$$

$$\begin{aligned} \mathbb{E}_{G_s}[V_Y] &= \frac{\mathbb{E}_{R_s}[V_Y V_S]}{\mathbb{E}_{R_s}[V_S]} && \text{[(18): } G_s = G_{R_s} \text{ reweighted by } v_S\text{]} \\ &= \mathbb{E}_{R_s}[V_Y] + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_S]} && \text{[}\mathbb{E}[XY] = \mathbb{E}X\mathbb{E}Y + \mathrm{Cov}(X,Y)\text{]} \\ \frac{\text{just treated}}{\text{not yet treated}} &= \delta(s \mid s, x)\Big(1 + \frac{\mathrm{Cov}_{R_s}(V_Y, V_S)}{\mathbb{E}_{R_s}[V_Y]\mathbb{E}_{R_s}[V_S]}\Big) && \text{[divide; } \lambda_Y\varphi_Y \text{ cancel]} \\ &= \delta(s \mid s, x) \cdot \frac{\mathcal{L}\,\mathcal{L}_{ab}}{\mathcal{L}_a\,\mathcal{L}_b}\Big|_{(a_s, b_s)} && \text{[survivor moments from Section 4.3]} \end{aligned}$$

- Observed ratio = causal effect × **selection factor**.
- Selection factor = 1 iff V_Y, V_S uncorrelated in the risk set: then timing is as good as random at s.
- The selection factor is known from Step A, so $\delta(s \mid s, x)$ is identified.

8.4 Step D: beyond the moment of treatment (Prop. 3)

→ *Why now:* Step C only gives δ at $t = s$. After s the treated cohort keeps being reshaped by exits, so we need its full survival curve.

For $y > s$, let $g(y, s \mid x)$ be the density of "treated at s and still unemployed at y", and $D := D(y \mid s, x)$:

$$\begin{aligned} g(y, s \mid x) &= \mathbb{E}\big[f_S(s \mid x, V)\, \Pr(Y > y \mid S = s, x, V)\big] && \text{[LIE; Assumption 2]} \\ &= \mathbb{E}\big[\lambda_S(s)\varphi_S V_S e^{-b_s V_S} \cdot e^{-\varphi_Y V_Y [\Lambda_Y(s) + D]}\big] && \text{[(3); (2) with (17)]} \\ &= \lambda_S(s)\varphi_S\, \mathbb{E}\big[V_S\, e^{-(a_s + \varphi_Y D)V_Y - b_s V_S}\big] && \text{[linearity; } a_s = \varphi_Y \Lambda_Y(s)\text{]} \\ &= -\lambda_S(s)\varphi_S(x)\, \mathcal{L}_b\big(a_s + \varphi_Y(x)D,\ b_s\big) && \text{[}\mathcal{L}_b = -\mathbb{E}[V_S e^{-aV_Y - bV_S}]\text{]} \end{aligned}$$

Divide by the same expression at $y \downarrow s$ (where $D = 0$), as in (16):

$$\begin{aligned} P_1(y \mid s, x) &= \frac{\mathcal{L}_b\big(a_s + \varphi_Y(x)D(y \mid s, x),\ b_s\big)}{\mathcal{L}_b(a_s, b_s)} && \text{[the } \lambda_S\varphi_S \text{ cancel]} \\ &= \mathbb{E}_{G_s}\big[e^{-\varphi_Y(x)V_Y D(y|s,x)}\big] && \text{[rewrite with weights (18)]} \end{aligned} \tag{19}$$

- Left side: data. Right side: everything known from Step A, except D.
- Strictly decreasing in $D \Rightarrow$ invert for $D(y \mid s, x)$.
- Then $\delta(y \mid s, x) = \partial_y D(y \mid s, x)/\lambda_Y(y)$ [fundamental theorem of calculus].
- This is the paper's eq. (10), divided by its value at $y \downarrow s$.

Two selection problems, both handled by knowing G

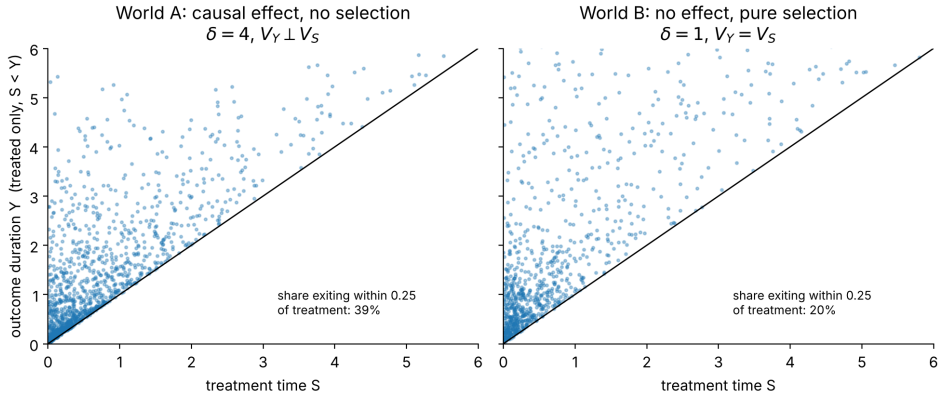
1. Selection *into* treatment at s: the v_S-tilt in (18).
2. Selection *out* after s: in (19), survivors of the treated cohort drift to low V_Y, as in Section 2.
   - Present whenever V_Y varies, even with random timing.
   - So naive hazard ratios at later y are not δ.

8.5 Summary: what is used where

→ *Why now:* collect which assumption did which job.

Assumption	Where it is used
No anticipation	Step A: before S there is no δ, so a clean competing-risks problem
Assumption 2	Steps A, B, D: independence given V; treated = risk set reweighted by v_S
MPH, $V \perp X$	Step A: everything is $\mathcal{L}$ of indices
$\mathbb{E}V < \infty$ (A4)	Step A, step 1: short-duration hazard ratios
Open set (A3)	Step A, step 2: trace $\mathcal{L}$ in 2-D $\Rightarrow$ dependence of V_Y, V_S

- Not used: an excluded instrument, selection on observables, parametric forms, multiple spells.



Intuition. Left: causal effect, no selection; exits bunch right after treatment at every S (local). Right: no effect, pure selection; S and Y correlated overall (global).